

CMOS Impedance Sensor with 128×128 Microelectrode Arrays for Measuring Surface Coverage of Microparticles

Hyunseo Shin¹, Dakyung Kang², Jungkyu Kim², and Nam-Seog Kim¹

¹School of Information and Communication Engineering, Chungbuk National University, Cheongju-Si, Republic of Korea

²Department of Mechanical Engineering, University of Utah, Salt Lake City, UT 84112, United States

E-mail: jkim@mech.utah.edu and namseog.kim@cbnu.ac.kr

Abstract— This paper presents a 128×128 CMOS microelectrode array for highly sensitive impedance biosensing applications. Conventional impedance sensing platforms demonstrated insufficient spatial resolution and poor signal-to-noise ratios, which limits detection sensitivity through bulk measurements. The proposed CMOS platform integrates 16,384 individually addressable microelectrodes with dedicated amplification circuitry, enabling a paradigm shift from bulk measurements to discrete microparticle counting. The system operates across a 20 kHz to 100 kHz frequency range with modular architecture comprising a CMOS microelectrode array chip, external microcontroller unit (MCU), and AD5933 impedance analyzer. Experimental validation demonstrates quantitative correlation between microparticle concentration and impedance change ($\Delta|Z|$) response across multiple orders of magnitude (1.02×10^6 to 1.02×10^9 microparticles/mL), with surface coverage ranging from 0.25% to 98.65%. A dual-measurement approach combining electrical characterization with optical microscopy validation confirms the accuracy of the sensor. The CMOS impedance sensor achieves single-microparticle detection capabilities with real-time spatial localization, establishing a foundation for digital biosensing at ultra-low concentrations suitable for point-of-care diagnostics and personalized medicine applications.

Keywords— *Electrochemical impedance spectroscopy, CMOS microelectrode array, single-microparticle detection, biosensor, point-of-care diagnostics*

I. INTRODUCTION

Highly sensitive biomarker detection is critical for early disease diagnosis [1]. However, traditional diagnostic methods often fail to detect biomarkers during early disease stages, missing the critical window when treatment is most effective [2]. A label-free impedance biosensor addresses this challenge by converting molecular binding events directly into measurable electronic signals [3]. Despite its simplicity and versatility, impedance biosensors face significant challenges in achieving sufficient detection sensitivity [4]. To address these challenges, researchers have focused on developing signal enhancement strategies by tweaking biorecognition elements. For example, microparticles or nanoparticles can significantly increase the sensitivity of impedance measurements by improving the capture efficiency of target analytes [5], [6] and altering impedance through changes in resistance or double layer capacitance [4], [7]. However, conventional interdigitated electrode (IDE) sensors fundamentally limit this strategy. The IDE sensors fabricated on passive glass or PCB substrates lack the spatial resolution and signal-to-noise ratios needed to detect individual microparticle binding events, which inherently limits sensitivity [8], [9].

Breaking through this sensitivity barrier demands a paradigm shift from bulk measurements to discrete

microparticle counting, achievable through high-density Complementary Metal-Oxide-Semiconductor (CMOS) microelectrode arrays (MEAs) [10], [11]. The integrated system, depicted in Fig. 1, combines a CMOS chip with a MEA and supporting electronics including signal input and readout interfaces [10]. The platform incorporates epoxy encapsulation for protection and a printed circuit board (PCB) for signal generation and readout interfacing.

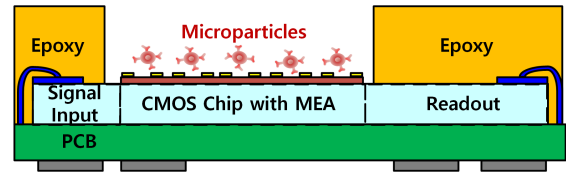


Fig. 1. Cross-sectional schematic of the CMOS impedance sensor system showing integrated microelectrode array (MEA) chip with protective epoxy encapsulation and PCB-based signal processing electronics for microparticle detection.

CMOS technology provides the distinctive advantage of monolithic integration – microscale electrodes are fabricated directly atop their dedicated amplification and processing circuitry [12]. This architecture dramatically enhances signal-to-noise ratio and enables fabrication of electrodes with dimensions precisely matched to single-microparticle. When a microparticle binds to one of these precisely scaled microelectrodes, it occupies a significant portion of the electrode area, inducing a large, unambiguous impedance change that is clearly distinguishable from background noise—a critical capability unachievable with traditional electrode arrays.

This paper presents the design, fabrication, and characterization of a high-density 128×128 CMOS MEA platform for single-microparticle detection. By leveraging 16,384 individually addressable microelectrodes, we demonstrate real-time detection and localization of discrete microparticle binding events. This work establishes a foundation for digital biosensing at ultra-low concentrations and sets a new benchmark for integrated electrochemical diagnostics.

II. SYSTEM ARCHITECTURE AND DESIGN

A. System Configuration

The CMOS impedance sensor adopts a modular architecture comprising three primary components: a custom CMOS chip housing the 128×128 microelectrode array, an external microcontroller unit (MCU) for system control and data management, and a dedicated analog front-end for precision impedance analysis as shown in Fig. 2. This distributed approach optimizes performance by leveraging the strengths of each component while maintaining system flexibility and scalability for different applications.

The CMOS chip serves as the core sensing platform, incorporating 16,384 individually addressable microelectrodes fabricated using standard CMOS processes. Each electrode is directly integrated with dedicated amplification and signal conditioning circuitry, enabling high signal-to-noise ratio measurements and precise localization of individual microparticle binding events. This monolithic integration approach eliminates parasitic effects and interconnection losses that typically limit discrete electrode arrays performance, while minimizing parasitic capacitances and reducing noise coupling.

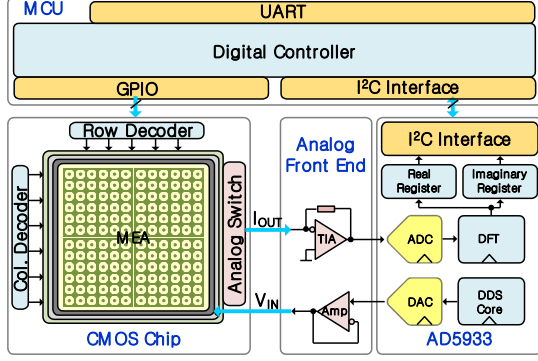


Fig. 2. System architecture of the CMOS impedance sensor featuring a 128×128 microelectrode array chip with integrated amplification, external microcontroller (MCU) for digital control, and AD5933 analog front-end for precision impedance analysis.

The MCU manages electrode selection and measurement sequencing via inter-integrated circuit (I²C) and general purpose input/output (GPIO), implements the data-acquisition protocol in its firmware, and streams formatted impedance data to the host PC over the universal asynchronous receiver/transmitter (UART). Row and column decoder circuits enable selective addressing of individual electrodes within the array, allowing both parallel and sequential measurement modes depending on application requirements. The digital controller implements real-time data processing algorithms for impedance calculation and microparticle detection, while managing the interface to external data storage and analysis systems. The analog front-end centers around the AD5933 impedance converter [13], incorporating a direct digital synthesizer (DDS) core for generating precise excitation signals, a precision analog-to-digital converter (ADC) for signal digitization, and a digital-to-analog converter (DAC) for bias control. Real and imaginary impedance registers provide direct access to complex impedance data, enabling efficient calculation of magnitude and phase information across the measurement frequency range. This modular design facilitates system optimization while maintaining compatibility with standard laboratory instrumentation and point-of-care diagnostic platforms.

B. Measurement Setup for CMOS Impedance Sensor Array

The measurement setup, illustrated in Fig. 3, employs an optical microscope system for real-time visualization and precision positioning of the CMOS chip. The system operates across 20 kHz to 100 kHz frequency range with the AD5933 impedance analyzer interfaced through a custom PCB that provides electrical connections, signal conditioning, and power management. Sample preparation involves controlled introduction of test solutions with varying microparticle concentrations through precision pipetting systems. Different microparticle concentrations (microparticles/mL) are tested to

validate the sensitivity of the sensor array and establish the correlation between microparticle surface coverage and impedance change ($\Delta|Z|$). Environmental controls ensure temperature stability and electromagnetic interference shielding, while calibration procedures using standard impedance references validate measurement accuracy across the frequency spectrum.

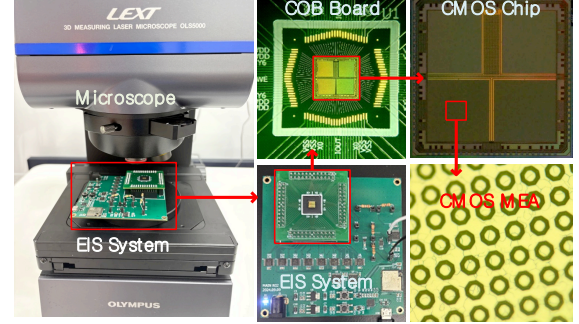


Fig. 3. Experimental measurement setup for CMOS impedance sensor characterization with optical microscopy serving as verification tool to validate electrical measurements of microparticle distribution on the MEA.

The experimental protocol incorporates microscope imaging to capture microparticle distribution patterns on the CMOS impedance sensors, providing visual validation for correlating optical observations with electrical measurements. Real-time data acquisition through the MCU interface enables continuous monitoring of $\Delta|Z|$ during microparticle binding events, allowing direct comparison between $\Delta|Z|$ and microscope images showing actual microparticle surface coverage on each electrode. This dual-measurement approach validates sensor performance by demonstrating quantitative agreement between microparticle coverage and impedance changes across different concentrations.

III. MEASUREMENT RESULTS

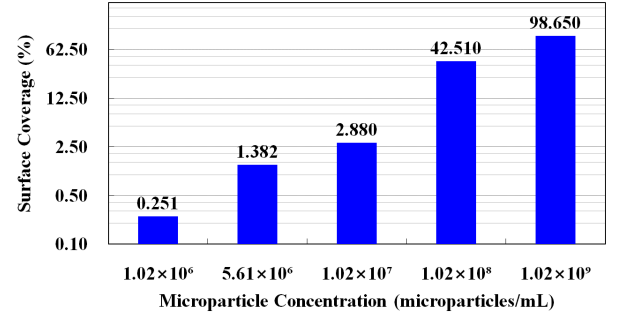


Fig. 4. Measured surface coverage percentage of microparticles as a function of concentration from 1.02×10^6 to 1.02×10^9 particles/mL.

Fig. 4 demonstrates the relationship between microparticle concentration and surface coverage percentage across the CMOS microelectrode array. The experimental results show a systematic increase in surface coverage from 0.25% to 98.65% as microparticle concentration increases from 1.02×10^6 to 1.02×10^9 microparticles/mL. This linear relationship validates the quantitative detection capability of the CMOS impedance sensor platform and establishes the operational range for reliable microparticle detection across multiple orders of magnitude in concentration.

Fig. 5(a) presents $\Delta|Z|$ for two representative microparticle concentrations from 20 kHz to 100 kHz. The measurements demonstrate clear differentiation between 1.02×10^8 and

1.02×10^9 microparticles/mL with $3 \mu\text{m}$ electrode spacing. Impedance magnitude decreases significantly with higher microparticle concentrations, confirming the sensor's ability to distinguish analyte levels through electrical impedance changes. The frequency-dependent response shows optimal sensitivity at lower frequencies, consistent with typical non-Faradaic impedance behavior. Fig. 5(b) illustrates electrode spacing impact on $\Delta|Z|$ measurements, comparing $3 \mu\text{m}$ and $4 \mu\text{m}$ spacings. The results demonstrate that reduced electrode spacing ($3 \mu\text{m}$) provides enhanced sensitivity with 1.63 times higher signal-to-noise ratio compared to larger spacing ($4 \mu\text{m}$) at 20 kHz. This validates the high-density electrode array design approach, where smaller electrode spacing enables better spatial resolution and increased sensitivity for single-microparticle detection applications.

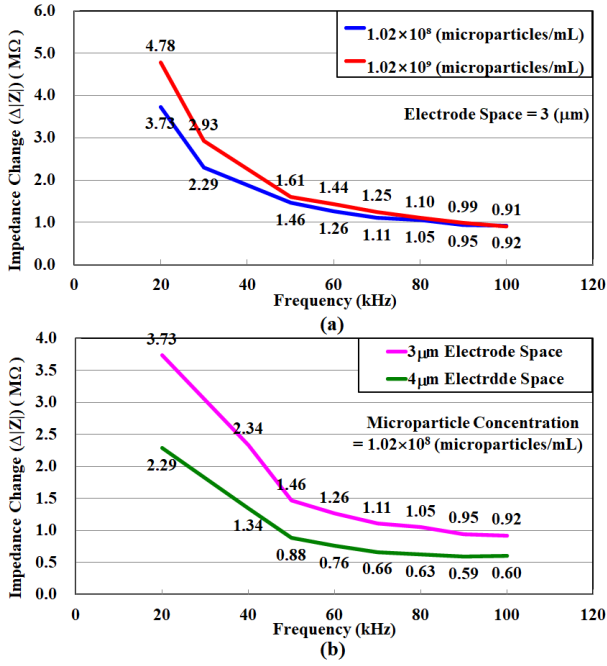


Fig. 5. Frequency-dependent impedance change ($\Delta|Z|$) for (a) the different microparticle concentrations and (b) the different electrode spaces.

Fig. 6 provides comprehensive validation through direct correlation between optical microscopy observations and electrical impedance heat maps. The microscopy images show microparticle distribution patterns across the CMOS sensor surface for different concentrations: 1.02×10^6 microparticles/mL in Fig. 6(a), 1.02×10^7 microparticles/mL in Fig. 6(b), and 1.02×10^8 microparticles/mL in Fig. 6(c), demonstrating single-microparticle detection capability. Furthermore, single-microparticle electrode analysis reveals $115.17 \text{ k}\Omega$ impedance change per microparticle. The corresponding impedance change heat maps demonstrate excellent spatial correlation with microparticle distributions, confirming the accuracy of electrical measurements in detecting and localizing individual microparticle binding events. The color-coded maps clearly show increased signal intensity with higher microparticle density, validating single-microparticle detection capability.

IV. CONCLUSION

This paper presents a novel high-density 128×128 CMOS microelectrode array platform for highly-sensitive impedance biosensing applications. The system demonstrates single-microparticle detection capabilities through 16,384

individually addressable microelectrodes, enabling real-time detection and spatial localization of discrete microparticle binding events. Experimental validation confirms a quantitative correlation between microparticle concentration and impedance change ($\Delta|Z|$) response across multiple orders of magnitude (1.02×10^6 to 1.02×10^9 microparticles/mL), with a dual-measurement approach establishing platform accuracy. This work establishes a foundation for digital biosensing at ultra-low concentrations, overcoming the sensitivity limitations of conventional approaches and setting a new benchmark for integrated electrochemical diagnostics with potential applications in point-of-care diagnostics and personalized medicine.

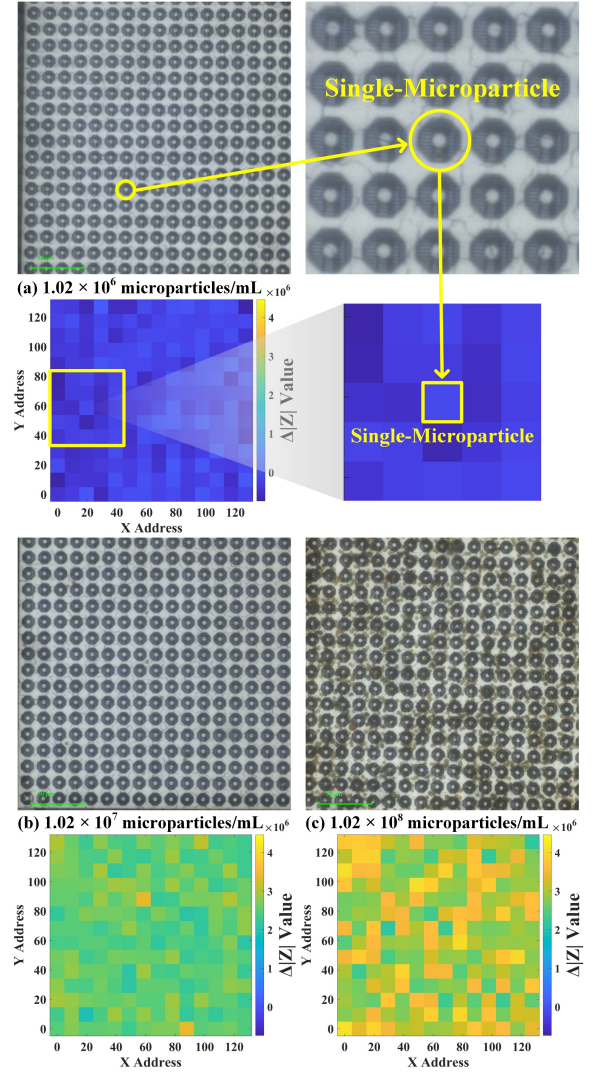


Fig. 6. Optical microscopy images (top) and corresponding two-dimensional impedance change ($\Delta|Z|$) heat maps from impedance sensor results (bottom) for microparticle concentrations of (a) 1.02×10^6 , (b) 1.02×10^7 , and (c) 1.02×10^8 particles/mL, demonstrating spatial correlation between particle distribution and electrical response.

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